

brAIIn

*A Multiclass Deep Learning System for Brain Tumor
Detection, Classification, and Interpretable Diagnostic Support*

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Note on this document. This is an English translation and consolidation of the project's original report ("memoria") and presentation slides, prepared for the project's public GitHub repository. It has been edited from the original bilingual (Catalan/Spanish) source in three ways: (1) translated in full to English; (2) the unsupervised anomaly-detection module (a VQ-VAE autoencoder), which was an independent contribution by a third collaborator not credited on this version, has been removed from the system description and results; (3) authorship is limited to the two contributors credited above. All classification, detection, uncertainty, interpretability, and clinical-narrative results reported here are unchanged from the original evaluation.

ABSTRACT

This project addresses the detection and multiclass classification of brain tumors in magnetic resonance imaging (MRI) through deep learning. The system builds on a previous binary classifier (tumor / no-tumor, ~99.3% accuracy) and extends the task to four classes: glioma, meningioma, pituitary tumor, and no tumor.

An ensemble of three convolutional neural networks — ConvNeXt-Tiny, EfficientNet-B3, and ResNet-50 — was trained on 14,095 unique axial MRI slices, combined from the BRISC-2025, Mendeley (Epic/CSCR Hospital), and Kaggle (Nickparvar) datasets after MD5 deduplication. The system incorporates five complementary contributions on top of the classifier: uncertainty estimation via MC Dropout (T=20), energy-based out-of-distribution detection, hierarchical visual explanations (Grad-CAM, Grad-CAM++, LayerCAM), supervised tumor localization with a YOLO11n detector trained on 3,064 radiologist-annotated MRIs, and clinical narrative generation with MedGemma 1.5 4B, served locally through Ollama.

On a held-out test set of 2,114 images, EfficientNet-B3 reaches 99.48% accuracy and a macro F1 of 99.46%, while the three-model soft-vote ensemble matches that accuracy with 99.39% three-way agreement. The YOLO11n detector reaches an mAP₅₀ of 0.918, reinforcing the system's ability to complement classification with visual tumor localization. A React frontend exposes the full pipeline — predictions, uncertainty, heat-maps, editable bounding boxes, a symptom-driven risk score, and a bilingual (English/Spanish) interface — for interactive review. These results indicate the system can produce consistent, interpretable predictions oriented toward assisted medical image analysis, though clinical integration would require prospective studies, broader external validation, and evaluation in real medical settings.

1. Introduction and Motivation

1.1 Clinical context

Brain tumors represent a significant challenge in medicine because of their heterogeneity, their potential neurological impact, and the need for accurate evaluation to guide diagnosis and treatment. Magnetic resonance imaging (MRI) has become a core technique for structural brain analysis, since it allows lesions, tumor masses, and anatomical alterations to be observed in fine detail. However, interpreting these images requires specialized expertise and can be affected by case complexity, image quality, or visual similarity between certain lesion types.

1.2 Importance of MRI

MRI is one of the principal imaging tools in neuro-oncology, particularly for its ability to differentiate soft tissue and provide detailed information. For brain tumors, it allows suspicious regions to be identified, their extent observed, and subsequent studies to be planned. Manual analysis of large image volumes, however, can be time-consuming and subject to inter-observer variability.

1.3 The role of AI in medical imaging

In recent years, deep learning techniques have taken on a prominent role in medical image analysis, largely because of their capacity to learn complex visual representations from large datasets. In tasks such as brain tumor detection and classification, convolutional neural networks can automate part of the analysis process and generate consistent predictions from previously labeled images. Even so, their use in medical contexts demands attention not only to predictive performance, but also to interpretability, uncertainty, and robustness.

1.4 Limitations of binary classifiers

As a starting point for this work, a binary classification system was developed previously, aimed at distinguishing between images with and without a tumor. It was built incrementally from a ResNet-50 architecture, progressively incorporating data augmentation, CLAHE contrast enhancement, noise reduction, deep-layer fine-tuning, test-time augmentation, and a lowered decision threshold. The final model reached approximately 99.3% accuracy and a 0.3% false-negative rate.

Although these results show strong performance at the general task of detecting the presence or absence of a tumor, this approach has important limitations when the goal is a more complete support tool. In a medical analysis context, it is not enough to indicate whether or not a suspicious lesion exists — it also matters to provide information about the likely tumor type, its

approximate location, the model's confidence, and which regions of the image influenced the decision. Binary classification is therefore a useful first approximation, but insufficient for this project's goals.

1.5 Interpretability and trust

In medicine, an AI system should not be limited to producing a final class label. For a support tool to be useful and understandable, it should also convey how confident its predictions are and which image regions influenced the decision. This project therefore incorporates uncertainty estimation, out-of-distribution detection, and visual explainability techniques, with the goal of increasing the transparency of the process.

1.6 Project motivation

The main motivation for this project comes from a combination of interest in AI applied to healthcare and the need to build systems that are more interpretable, safer, and more useful for diagnostic-support tasks. From a technical perspective, the project integrates several areas of deep learning — image classification, uncertainty estimation, anomaly/out-of-distribution detection, visual explainability, object localization, and clinical text generation. From an applied perspective, the system aims to approximate a more complete workflow, in which an automatic prediction is accompanied by visual information that facilitates its interpretation.

2. Objectives

2.1 General objective

The objective of this project is to develop an intelligent support system for the analysis of brain MRI images through deep learning techniques, oriented toward the detection and multiclass classification of brain tumors. The system extends a previous binary classifier into a more complete solution, capable of differentiating between several tumor categories while incorporating complementary mechanisms for visual interpretability, uncertainty estimation, out-of-distribution detection, supervised tumor localization, and interactive visualization of results. The project does not aim to replace the judgment of a medical specialist, but to build an experimental tool that facilitates the understanding, review, and analysis of predictions generated by AI models applied to medical imaging.

2.2 Specific objectives

- Extend the previous binary classification system toward a multiclass approach, differentiating between glioma, meningioma, no tumor, and pituitary tumor from brain MRI images.

- Implement and train different convolutional neural network architectures — specifically ConvNeXt-Tiny, EfficientNet-B3, and ResNet-50 — to compare their classification performance.
- Evaluate model performance using quantitative metrics such as precision, recall, F1-score, and macro F1, paying particular attention to errors relevant in a clinical context.
- Incorporate uncertainty-estimation mechanisms, using techniques such as MC Dropout, to obtain an orientative measure of prediction reliability.
- Integrate an out-of-distribution detection system to identify images that may fall outside the training domain and therefore produce less reliable predictions.
- Apply visual interpretability techniques — Grad-CAM, Grad-CAM++, and LayerCAM — to represent the image regions with the greatest influence on the model's decision.
- Complement classification with visual tumor localization, using a YOLO11n detector capable of marking suspicious regions with bounding boxes on the MRI images.
- Develop an interactive web interface in React that allows the analysis flow to be visualized, generated predictions to be reviewed, heat-maps to be inspected, bounding boxes to be corrected, and overall interaction with the system.
- Incorporate a clinical-narrative generation layer, based on a locally deployed language model, to contextualize results in a more understandable and structured way.
- Analyze the technical, methodological, and clinical limitations of the developed system, identifying possible lines of improvement related to external validation, data diversity, model generalization, and future evaluation in real medical settings.

3. Related Work

In recent years, the application of intelligent systems to medical neuroimaging has advanced significantly, especially in the analysis of brain MRIs for tumor detection, segmentation, classification, and monitoring. According to Dorfner et al. (2025), deep learning models have taken on a relevant role in brain tumor analysis, as they can automate complex tasks, generate more objective and reproducible measurements, and support clinical processes related to diagnosis, treatment planning, and disease monitoring.

This progress is particularly important in neuro-oncology, where precise lesion delineation, differentiation between tumor types, and reduced diagnostic variability are fundamental to improving clinical decision-making.

One of the most relevant international benchmarks in this field is the BraTS challenge (Brain Tumor Segmentation Challenge), active since 2012 and oriented toward the comparative evaluation of machine learning methods applied to brain tumor segmentation in MRI. Its importance lies in having provided annotated datasets, common metrics, and standardized evaluation scenarios, allowing objective comparison of different models' performance. BraTS's

recent evolution shows a progressive expansion of its scope: the 2025 edition covers multiple tasks related to gliomas, meningiomas, brain metastases, and pre- and post-treatment scenarios.

Within the Spanish clinical research space, DISCERN stands out — a tool developed by researchers affiliated with Vall d'Hebron, VHIO (Vall d'Hebron Institute of Oncology), IDIBELL (Bellvitge Biomedical Research Institute), and other collaborating centers. This system uses information obtained from MRI to support the non-invasive differential diagnosis of brain tumors. Unlike approaches focused solely on detecting the presence or absence of a lesion, DISCERN differentiates between three common types of malignant brain tumors, reaching 78% accuracy on that classification — a useful reference point for how difficult multiclass differentiation from imaging alone actually is.

Another notable line of work is the algorithm developed by the Universidad Politécnica de Madrid and CIBER-BBN, in collaboration with Children's National Hospital in Washington. This work focuses on the detection and segmentation of brain tumors, particularly gliomas, via MRI, and placed first in the Brain Tumor Segmentation Challenge.

The current state of the field shows an evolution from models focused solely on prediction toward more interpretable, clinically-oriented systems. In medical settings, obtaining a correct classification is not enough — it is also necessary to visually justify which regions of the image influenced the system's decision. In this sense, techniques such as Grad-CAM and Grad-CAM++ have been widely used to generate activation maps that approximate the location of relevant regions within an image, increasing model transparency and facilitating specialist review.

Nonetheless, clinical adoption of these solutions still faces significant challenges. Although AI-based medical devices authorized for commercial use already exist, bodies such as the FDA emphasize the need to guarantee a system's safety, efficacy, transparency, and fitness for its intended use before clinical integration. Recent reviews highlight that many research models still rely on limited or narrow datasets, which can hinder generalization to real clinical scenarios.

4. Technologies and Techniques

4.1 Image preprocessing

This section groups the techniques applied before images are processed by the model. Their main goal is to improve the visual quality of the MRI scans and help the model learn more representative visual patterns, reducing the influence of artifacts or irrelevant variation.

Noise reduction via Gaussian filtering

A Gaussian filter is used to smooth MRI images and reduce high-frequency noise, decreasing the influence of irrelevant variation that could affect model learning. Its application must be tuned carefully to avoid losing genuine detail.

Adaptive contrast enhancement via CLAHE

CLAHE (Contrast-Limited Adaptive Histogram Equalization) improves contrast locally, adapting intensity across different regions of the image. This is especially useful in MRI, where regions of very different intensity can coexist — it can improve the visibility of regions of interest without over-exposing the brighter areas.

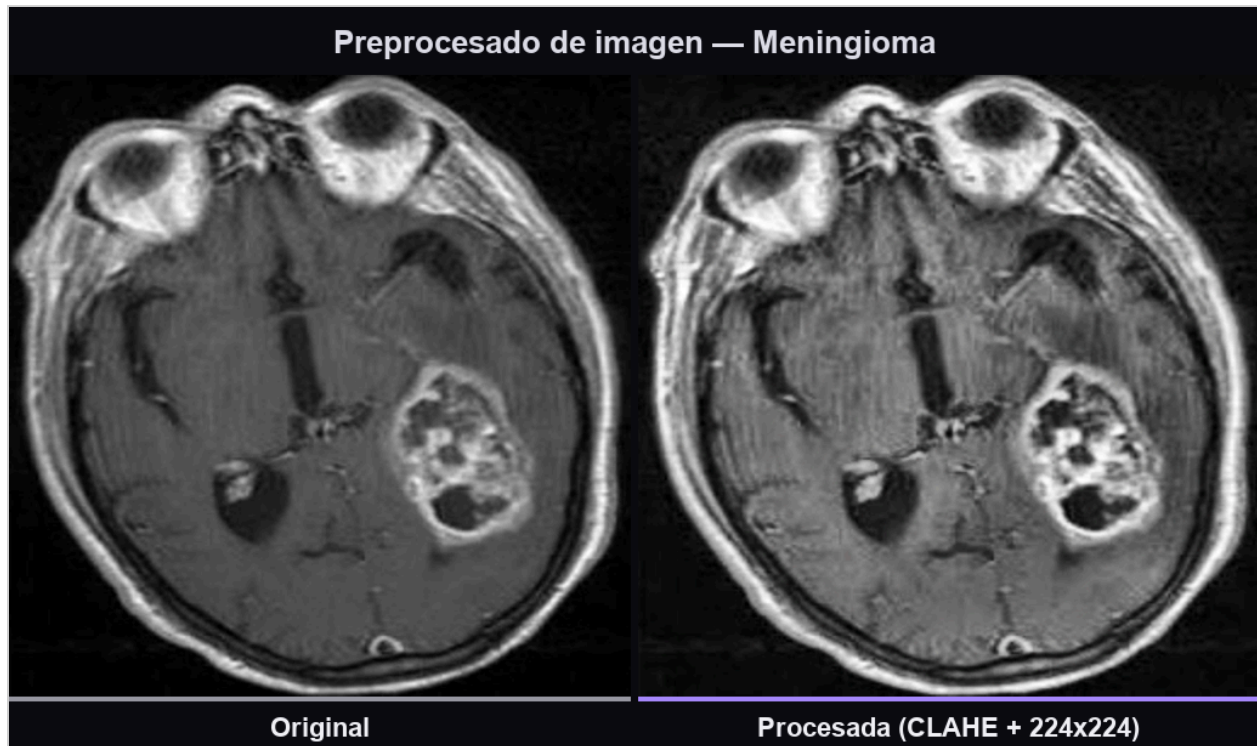


Figure 1. CLAHE applied to a real meningioma scan — before (left) and after (right). The technique visibly recovers soft-tissue contrast that a flat histogram leaves washed out.

Data augmentation

Data augmentation generates new variants of the original images through controlled transformations, such as rotations of up to $\pm 15^\circ$, horizontal flips, brightness/contrast adjustment, CLAHE, mild Gaussian noise, and a light elastic transform. These transformations are applied moderately, to preserve anatomical coherence and avoid generating unrealistic examples. Every one of the 9,867 training images is re-augmented independently on the fly, every epoch — nothing augmented is written to disk. Over a 40-epoch run this produces **394,680 distinct stochastic views** of the training set ($9,867 \times 40$), averaging roughly 2.5 transformations applied per view.

Transform	Probability	Applications / epoch	Total over 40 epochs	Rationale
Horizontal flip	0.50	$\approx 4,934$	$\approx 197,340$	Tumor type doesn't depend on left/right side; laterality comes

				from the bounding box
Rotation $\pm 15^\circ$	0.50	$\approx 4,934$	$\approx 197,340$	Realistic range; larger angles break radiological convention
Brightness / contrast $\pm 20\%$	0.50	$\approx 4,934$	$\approx 197,340$	Robustness to exposure differences between acquisitions
CLAHE	0.50	$\approx 4,934$	$\approx 197,340$	Matches the inference-time preprocessing exactly
Gaussian noise ($\sigma=0.01$)	0.30	$\approx 2,960$	$\approx 118,404$	Simulates acquisition noise without erasing real detail
Elastic transform (mild)	0.20	$\approx 1,973$	$\approx 78,936$	Simulates inter-scanner geometric variability

Table 1. Data augmentation transforms applied during training, with per-epoch and 40-epoch application counts. Deliberately not applied: vertical flip (the superior/inferior anatomical direction is meaningful, unlike left/right) and strong color perturbation (T1 MRI is essentially grayscale). Source: `src/train_v2.py` · `build_train_transform()`.

4.2 Model optimization

This section covers techniques applied during training and inference to improve performance, reduce critical errors, and increase generalization.

Fine-tuning of deep layers

Fine-tuning adjusts intermediate and deep layers of the model, rather than training only the final layers. This helps adapt previously learned representations to the specific characteristics of brain MRI, allowing the model to learn patterns better suited to tumor classification.

Class-weight adjustment

This technique assigns a higher weight to the tumor class during training, so the model penalizes errors on tumor-containing images more heavily. As a consequence, the system's sensitivity increases and the number of false negatives is reduced.

Test-Time Augmentation

Test-Time Augmentation (TTA) performs several predictions on the same image at inference time, applying different controlled transformations — the original image, a horizontal flip, and a vertical flip — then averages the resulting probabilities. This produces a more stable prediction and reduces dependence on a single input orientation.

Classification threshold adjustment

The classification threshold was lowered from 0.5 to 0.3: the model classifies an image as containing a tumor if the predicted probability exceeds 30%. This increases system sensitivity and reduces the risk of false negatives, at the cost of potentially more false positives — a deliberate choice to prioritize catching possible positive cases in a medical-support context.

4.3 Visual interpretability

This section covers the techniques used to visually explain the model's decisions. Their purpose is not only to classify the image, but to show which regions most influenced the prediction, making the system's behavior more transparent.

Grad-CAM++

Grad-CAM++ generates heat-maps indicating which regions of the image contributed most to the model's prediction, visually identifying the zones most relevant to the network's decision.

Heat-map overlay

The system produces a composite visualization: the original MRI image alongside the same image with the heat-map overlaid, facilitating visual interpretation of the contributing regions — though this should not be understood as an automatic clinical confirmation of tumor location.

Heat-map color coding

The heat-map uses an intuitive color scheme to represent each region's degree of activation or contribution to the model's prediction, intended as a visual aid for understanding system behavior rather than a clinical measurement.

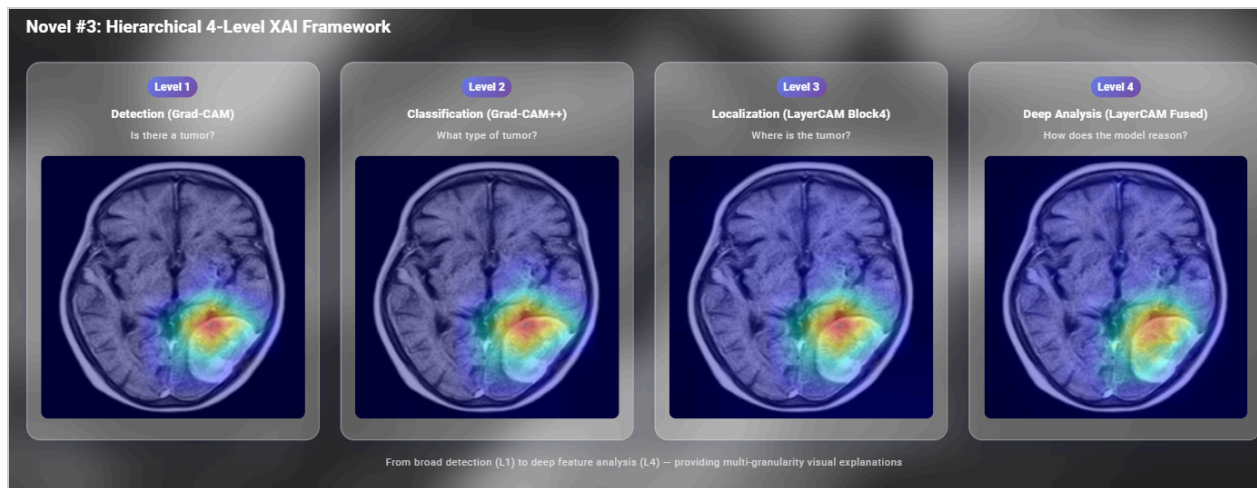


Figure 2. Hierarchical 4-level explainability: Grad-CAM (is there a tumor?), Grad-CAM++ (what type?), LayerCAM block4 (where exactly?), and fused LayerCAM (how does the model's reasoning combine across layers?). All four levels consistently highlight the same lesion.

5. Project Organization

The project was developed incrementally, following an organization based on successive phases: data preparation, model training, experimental evaluation, integration of complementary modules, and interface development. This approach made it possible to start from a prior binary system and evolve toward a more complete multiclass solution.

This organization allowed each component to be validated progressively before integration into the final system. First, the combined dataset was consolidated and the classification models were trained. Next, modules aimed at improving the reliability and interpretability of predictions were added. Finally, the results were integrated into a web interface designed to facilitate consultation, review, and understanding of the full workflow.

6. Methodology

6.1 Datasets

Before selecting the final datasets, several candidate sources were evaluated against criteria such as provenance, clinical validation, and image characteristics. While the datasets analyzed are useful in academic and experimental contexts, the lack of clinical validation limits their use in real medical settings.

Together, these datasets allow for a more complete, diverse brain tumor detection system, appropriate to the project's goals. Unlike the binary datasets used previously, these sets include images labeled across several categories — glioma, meningioma, pituitary tumor, and no tumor

— extending the model's scope from simple tumor/no-tumor detection to more precise classification.

The Mendeley dataset contributes a large volume of images organized into four classes, useful for training on a wider variety of cases. BRISC-2025 adds value beyond traditional datasets since, in addition to multiclass labels, it includes tumor-segmentation information and is annotated by experts for both classification and segmentation tasks. Finally, the Nickparvar (Kaggle) dataset was selected as one of the most widely used brain-tumor classification sets on Kaggle; its multiclass structure facilitates comparison with other academic work.

In the latest version of the system, these three datasets were consolidated into a single combined set. An MD5-based deduplication process discarded 11,169 duplicates — approximately 44% of the initial pool. After deduplication, the combined set contains 14,095 images. A stratified, per-class split of 70% train / 15% validation / 15% test follows. The test partition (2,114 images) is not used during any training or tuning phase, allowing an independent evaluation of the final system's performance.

	Total	Glioma	Meningioma	No Tumor	Pituitary
Raw (3 datasets)	25,264	6,974	6,164	5,439	6,687
After MD5 dedup	14,095	4,011	2,821	4,121	3,142
Train · 70%	9,867	2,808	1,975	2,885	2,199
Validation · 15%	2,114	602	423	618	471
Test · 15%	2,114	601	423	618	472

Table 2. Dataset consolidation: 3 classification datasets (BRISC-2025 · Mendeley · Nickparvar) combined and deduplicated by MD5 hash, then split 70/15/15 (stratified by class). The test partition is never touched during training or tuning. A separate dataset by Jun Cheng (Figshare, 3,064 radiologist-annotated MRIs, patient-disjoint split) contributes the region annotations used to train the YOLO11n detector.

6.2 System architecture

The system follows a client-server architecture, with a Flask backend acting as the processing core. On startup, the server loads all classification, localization, and interpretation models into memory, avoiding repeated initialization and reducing response times. The backend exposes a REST API through which the frontend sends images, requests predictions, queries metrics, and retrieves heat-maps and tumor-localization information.

The React frontend is served as a set of static files directly from the Flask server, simplifying deployment by integrating the visual application and backend within a single structure. In parallel, Ollama runs alongside the backend with MedGemma 1.5 4B loaded locally on CPU, used to generate clinical narrative from the system's results.

6.3 Flask backend and REST API

The backend is the system's central integration layer. Its main role is to receive requests from the frontend, process input images, and coordinate execution across the different inference modules — maintaining a clear separation between the visual interface, the processing layer, and the deep learning models. Flask handles requests related to prediction, metrics, heat-map generation, tumor localization, manual bounding-box editing, and clinical-narrative generation, acting as the intermediary between the user and the system's various intelligent components. Some results are returned in structured form so the frontend can render them visually — heat-maps are serialized as encoded images, and prediction responses include the estimated class, confidence, uncertainty, localization, and clinical context.

6.4 React frontend

The frontend serves as the presentation and interaction layer. Its main purpose is to facilitate image upload, display the backend's results, and allow visual review of the complete process. From this interface, the user can consult the predicted class, system confidence, heat-maps, bounding boxes, clinical context, and the model's key metrics. The application is organized into three main areas: an Analysis area for uploading images and viewing results, a Metrics area for reviewing aggregated performance, and an area for visualizing the system's overall pipeline. The frontend also supports English and Spanish. The interface functions not only as a visual component but as a tool for consultation and review of the system.

6.5 Interface components and associated models

The interface is organized into three main areas. The *Analysis* tab concentrates the primary interaction: image upload, running the analysis, viewing the prediction, reviewing localization, and consulting the clinical narrative. The *Metrics* tab presents experimental results and model performance. Finally, the *How It Works* tab offers a visual explanation of the complete system flow, from the MRI scan to the generated report.

Prediction and model comparison

The prediction header shows the estimated class and top-line confidence, drawn from the best-performing model among the three trained classifiers. A complementary Model Comparison component shows one row per model. When all three agree, the system presents a more stable result; when they disagree, the interface surfaces review signals, helping the user identify predictions that warrant more cautious interpretation.

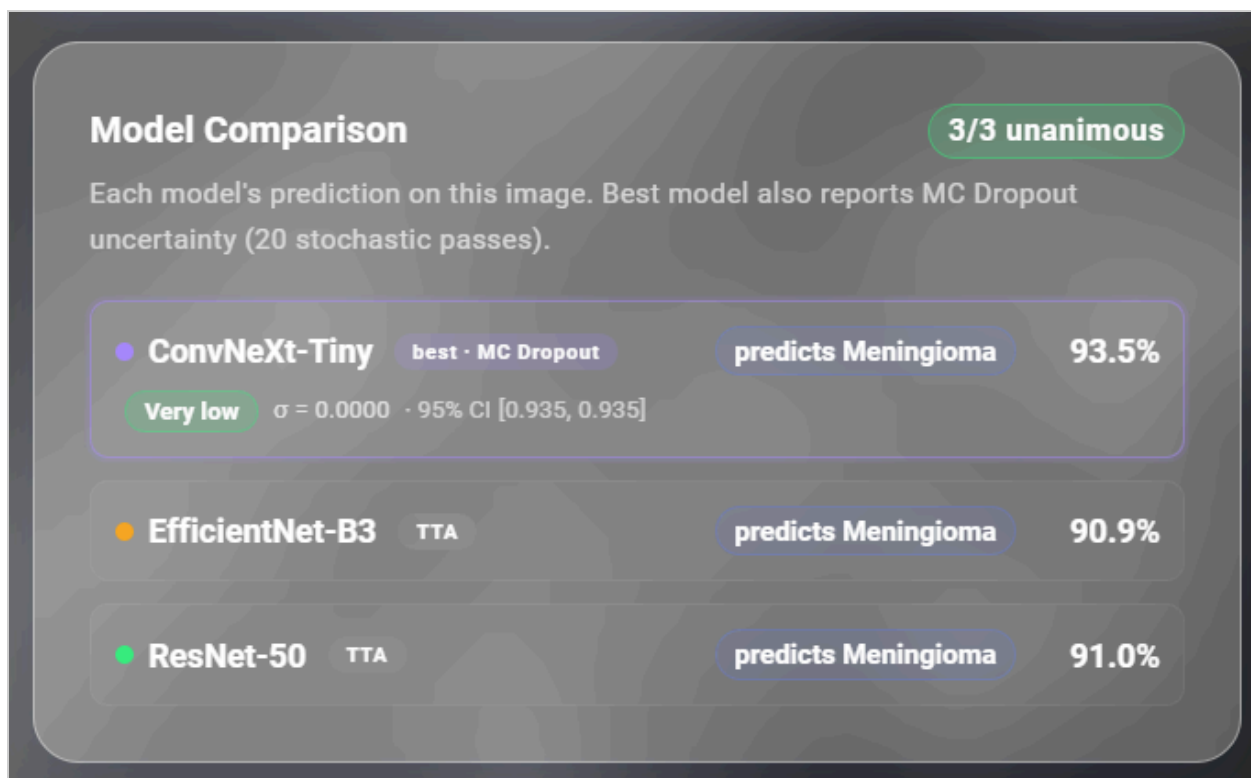


Figure 3. Upload panel and three-model ensemble agreement view. Every scan is classified independently by all three CNNs; a unanimous result is shown as such.

Reliability, uncertainty, and out-of-distribution detection

A Trust Verdict component acts as a visual reliability summary, combining several backend-computed signals: MC Dropout uncertainty, out-of-distribution detection, agreement between classifiers, and consistency of the analysis over the region of interest. The result is shown as a simple visual indicator, designed for a quick read of overall analysis status. An Uncertainty card shows more specific information about model variability — the system runs multiple stochastic passes over the same image and computes an orientative uncertainty measure. An OOD banner warns when the image has characteristics that may fall outside the training domain, helping to contextualize less reliable predictions.

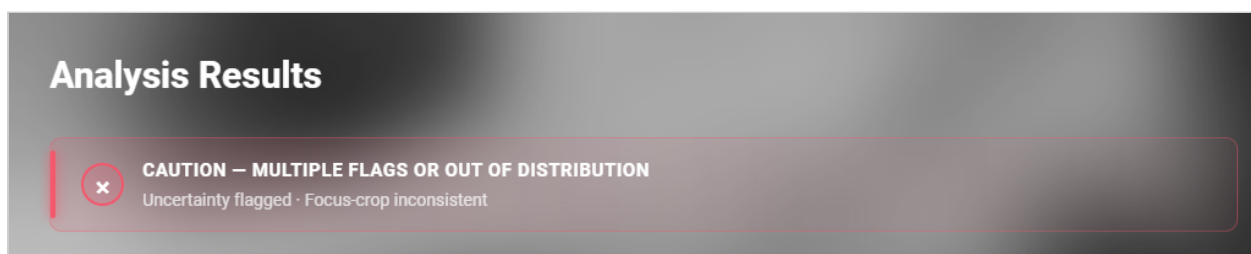


Figure 4. Trust-verdict banner flagging a scan for review, even with a confident, unanimous classification — independent consistency checks can still trip.

Visual interpretability and localization of the suspicious region

Visual interpretability components present heat-maps generated via Grad-CAM, Grad-CAM++, and LayerCAM. These maps should not be interpreted as a clinical segmentation, but as an indication of the regions that most influenced the model's decision, making classifier behavior more transparent and easier to review. Visual localization is complemented by a YOLO11n detector and MobileSAM: YOLO11n provides an initial bounding box for the suspicious region, and MobileSAM can refine that region with a segmentation mask. This information is shown in the Malignancy card as a crop of the image with the bounding box overlaid, allowing visual review of the region used to estimate size, location, and the orientative risk score.

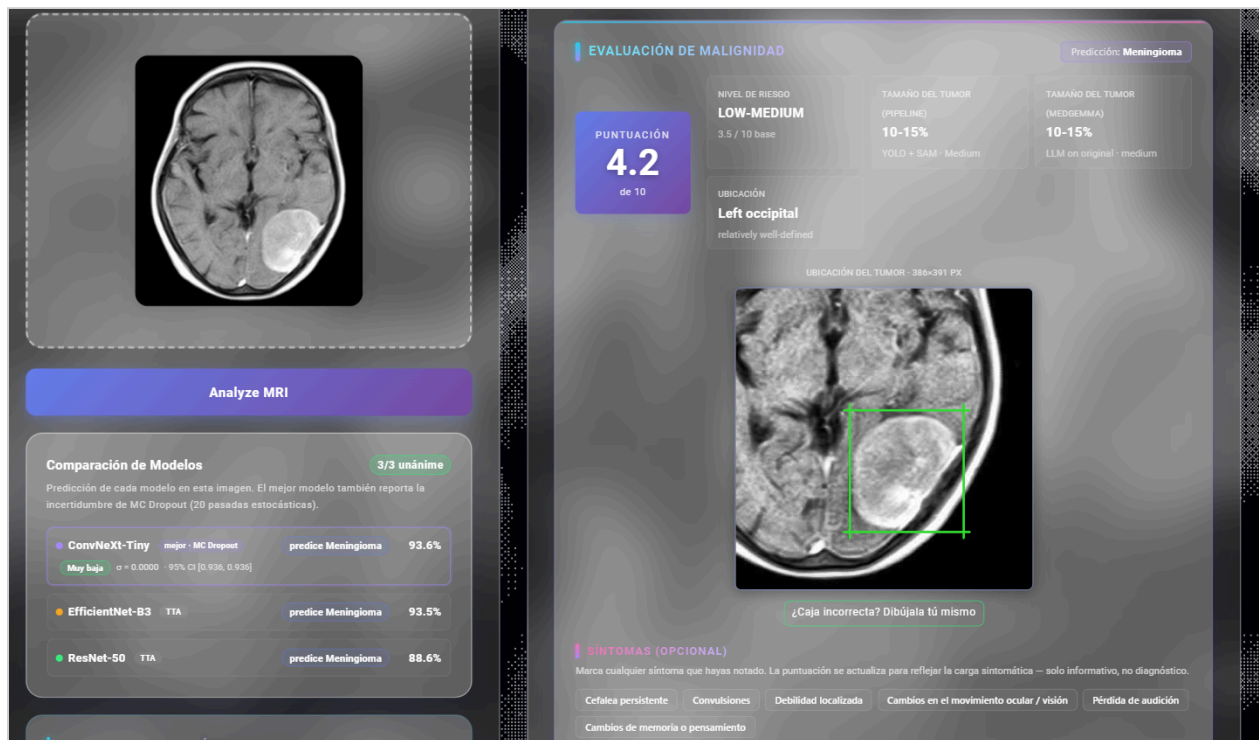


Figure 5. Malignancy assessment card: predicted class, tumor bounding box, and two independent size estimates — the pixel-based YOLO11n + MobileSAM pipeline, and MedGemma's own visual estimate — shown side by side so any disagreement stays visible.

Orientative score, clinical context, and narrative

The Malignancy card summarizes information from several subsystems: a base risk associated with the predicted class, the estimated size of the suspicious region, its approximate anatomical location, and model confidence. From these elements, the system generates an orientative risk score, to be interpreted as an experimental support indicator, not a diagnosis. The Clinical Context panel combines static reference information with dynamic fields generated by MedGemma 1.5, producing a more structured explanation of the predicted class. The Diagnostic Impression component uses MedGemma 1.5 4B (via Ollama) to generate a textual report in Basic or Advanced mode, adapting the level of detail to the type of user.

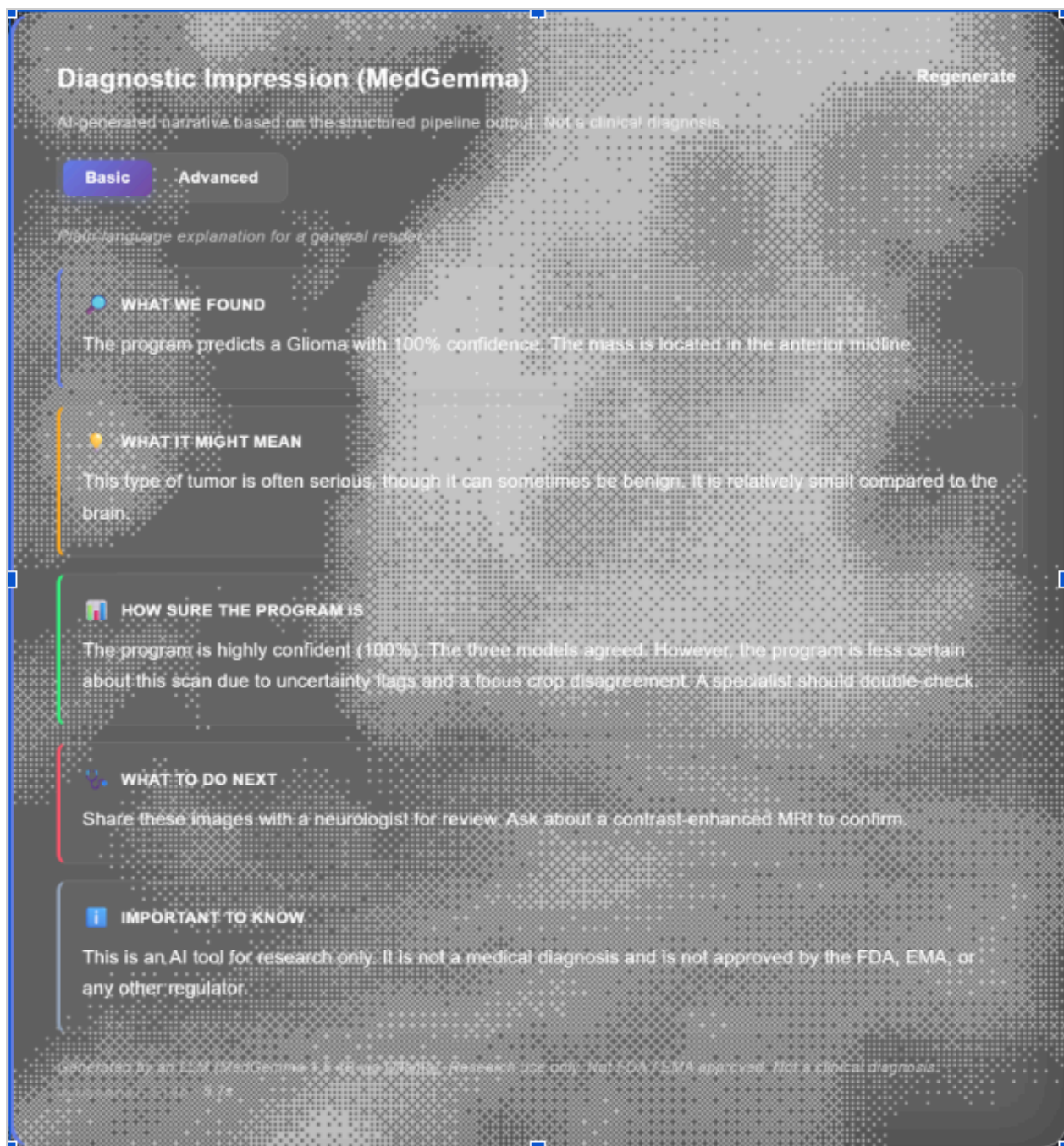


Figure 6. MedGemma-generated diagnostic report, combining static clinical reference text with dynamic fields produced by the model from the current prediction.

User interaction components

The interface includes elements that let the user interact with the result rather than only observe a closed prediction. A symptom selector adjusts the orientative score based on symptoms the user marks, using local rules rather than a separate clinical prediction model. Manual bounding-box editing allows the suspicious region to be corrected when automatic detection is not well aligned — the frontend sends the corrected coordinates to the backend, which re-analyzes the selected crop. Additional supporting elements include suggested questions for the specialist and alternative anatomical color-map views of the same image.



Figure 7. The same MRI slice re-rendered in several colormaps (HOT, JET, BONE, VIRIDIS, and others), which make different tissue contrasts visible that a single grayscale view can hide.

6.6 Metrics page

The Metrics page allows the models' experimental performance to be reviewed and compared from several angles. Unlike the Analysis tab, this section does not analyze a specific image, but represents aggregated results from evaluating the models on the test set. It is especially useful for justifying the choice of primary model, since it exposes not only final accuracy but also inference cost, model size, per-class performance, and error patterns visible in the confusion matrices.

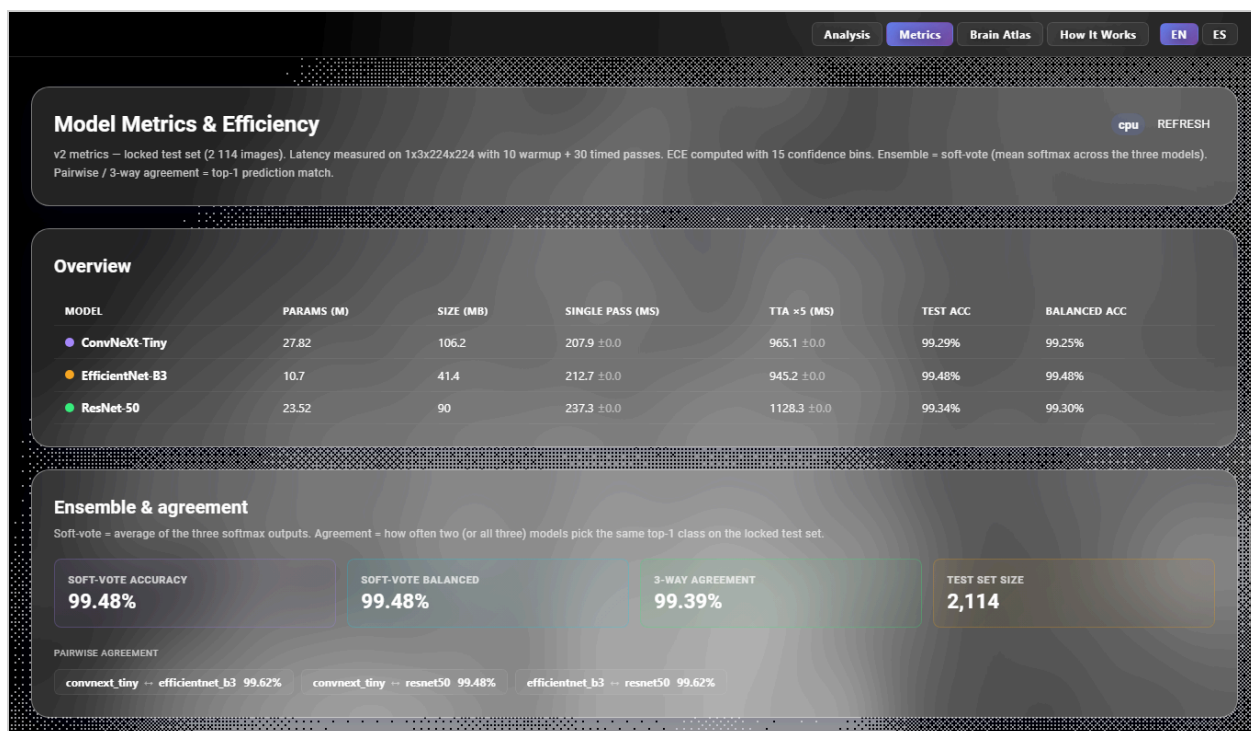


Figure 8. Metrics page: per-model overview (parameters, size, latency, accuracy) and ensemble agreement statistics, captured live from the running application.

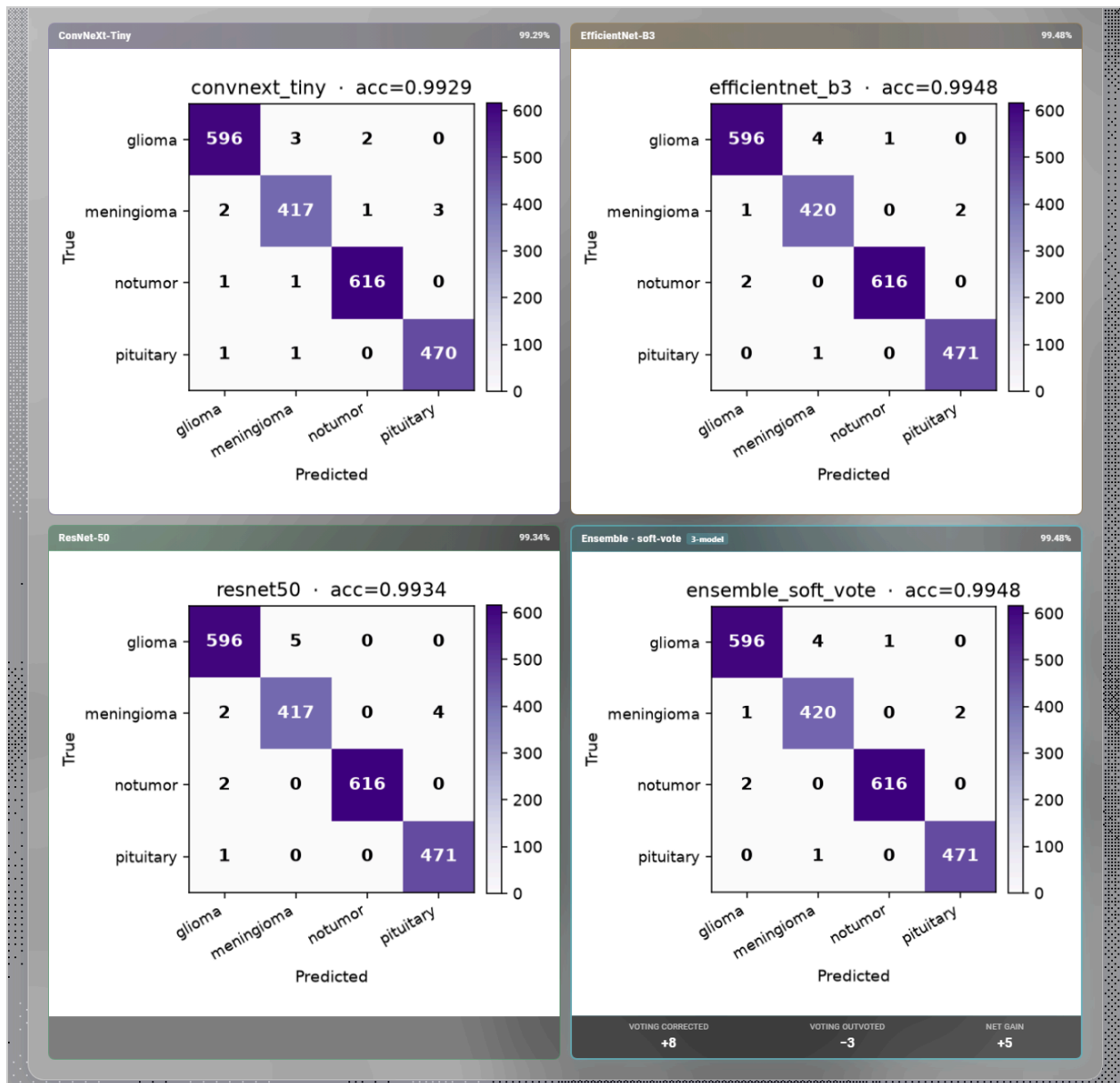


Figure 9. Per-model and soft-vote-ensemble confusion matrices on the locked 2,114-image test set, captured live from the running application.

6.7 How It Works page

The How It Works page serves an explanatory function. It does not run models in real time, but visually represents the system's stages, helping the user understand how the different modules connect within the analysis flow.

6.8 Inference modules

The architecture is designed modularly, so each component fulfills a specific role within the flow. This separation eases maintenance and allows individual modules to be extended or replaced without restructuring the whole application.

The first module handles multiclass classification, assigning the MRI image to one of the project's defined categories. Complementary modules are layered on top of this prediction to improve reliability and interpretability. Second, the system integrates an uncertainty-estimation module based on MC Dropout, running several inferences over the same image to obtain an averaged prediction and an orientative measure of model variability. An out-of-distribution detection module is also incorporated, aimed at identifying images that may fall outside the domain learned during training — especially relevant in medical contexts, where an atypical or low-quality input could produce a less reliable prediction. Finally, the system includes visual interpretability and tumor localization modules: interpretability techniques generate heat-maps showing which image regions most influenced the prediction, while supervised localization marks suspicious regions with bounding boxes.

Tumor localization uses a YOLO11n detector trained specifically to identify suspicious regions in MRI images. When the detector produces a sufficiently confident prediction, the resulting bounding box feeds into a more precise segmentation step via MobileSAM. If this process does not yield an adequate result, the system can fall back to alternative methods based on activation and intensity maps. This combination allows the system to provide not just a final class, but also a spatial reference for the possible region of interest — though these outputs should be interpreted as visual support, not an automatic confirmation of tumor location.

6.9 Ollama and MedGemma integration

Clinical narrative generation is implemented through Ollama, running as a service parallel to the main backend. MedGemma 1.5 4B operates within this environment, generating structured textual explanations from the system's results. Flask communicates with Ollama via an API, sending relevant information such as the predicted class, confidence level, estimated location, and other data derived from the image analysis. From this information, MedGemma generates a textual response oriented toward contextualizing the results and making them more understandable.

6.10 General operation flow

The system's operation begins when the user uploads an MRI image through the React interface. The frontend sends the image to the backend via a REST API request. Once received, the backend prepares the image and runs the multiclass classification models. After obtaining the primary prediction, the system computes complementary information related to confidence, uncertainty, and possible out-of-distribution detection. In parallel, heat-maps are generated to visually explain the model's decision, and the localization module runs to obtain a bounding box or mask for the suspicious region. Flask can then request clinical-narrative generation from Ollama via MedGemma 1.5 4B. Finally, the backend returns a structured response including the predicted class, confidence, heat-maps, visual localization, clinical text, and other supporting elements. The interface presents these results visually and interactively for review.

6.11 Brain Atlas page

The Brain Atlas page incorporates an interactive three-dimensional representation of the human brain that serves as anatomical reference alongside the MRI analysis. Its purpose is not to replace the analyzed image or offer an independent clinical interpretation, but to help the user visually contextualize the region detected by the system and understand which cognitive, motor, or sensory functions are typically associated with each brain region. The brain is rendered translucently and divided into eleven main anatomical regions: the frontal, parietal, temporal, and occipital lobes of both hemispheres, the cerebellum, the brainstem, and the pituitary gland. The user can rotate, zoom, and hover over any region to highlight it, at which point the system shows a brief description of the functions associated with that area. To connect the automatic prediction with the Atlas's anatomical information, the report generated by MedGemma includes a clickable element that directs the user to the region detected by the system — for example, an output such as "View this area in the Brain Atlas · Left Temporal Lobe" links the clinical narrative to an interactive visual representation of the mentioned area.

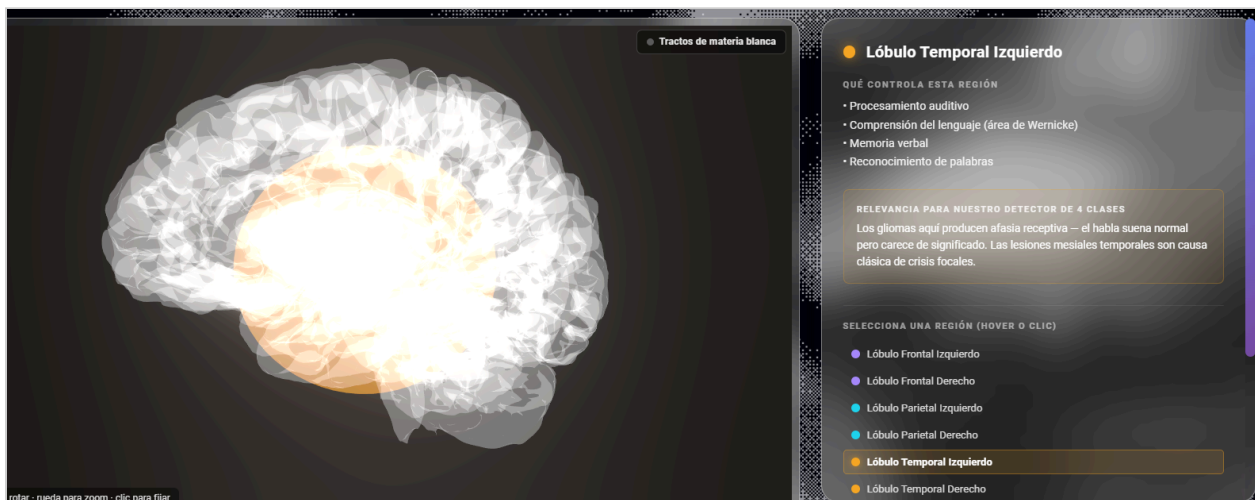


Figure 10. Interactive 3D Brain Atlas, showing a selected region with associated functional information.

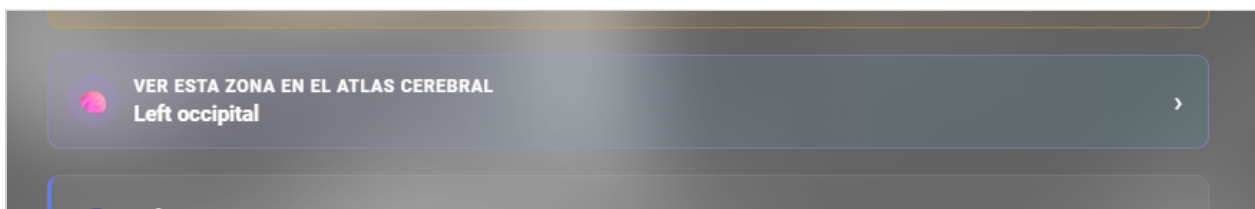


Figure 11. Clickable link from a MedGemma-generated report to the corresponding region in the Brain Atlas.

7. Evaluation

7.1 Protocol

System evaluation is organized across three levels, all applied to data not used during the final model's training. This separation allows the system's performance to be analyzed at different stages and reduces the risk of results being influenced by data already seen during development. First, a validation set is used during training to apply strategies such as early stopping, tune hyperparameters, and select training configurations in a controlled way. Second, a held-out test partition from the original split is preserved — not used during training or tuning — allowing an independent evaluation of the system's performance. Finally, a combined test set is used for the final, complete evaluation, allowing model behavior to be assessed on a broader, more representative distribution drawn from the project's combined dataset.

7.2 Metrics computed

To evaluate the classification models, the evaluation process runs on the combined test set. All three primary models are executed and a confusion matrix is computed, allowing both overall performance and class-specific errors to be analyzed. From this matrix, per-class metrics are derived — precision, recall, F1-score, and support. Precision measures the proportion of correct positive predictions for a given class, recall measures the model's ability to correctly identify samples belonging to that class, and F1-score combines the two via a harmonic mean, offering a balanced measure of performance when class sizes differ. Support indicates the number of real samples available for each category. In addition to per-class metrics, global metrics are computed — accuracy, macro precision, macro recall, macro F1-score, and their weighted equivalents. Macro metrics average without regard to per-class sample count, capturing balanced behavior across categories, while weighted metrics account for each class's support, giving a view more proportional to each category's actual weight in the evaluation set. Results are stored in JSON files, later loaded to be displayed within the system's interface.

7.3 YOLO detector evaluation

The YOLO11n detector is evaluated on the validation subset of the Cheng dataset, which is patient-disjoint — no patient used in validation also appears in training — reducing the risk of overlap between similar samples and yielding a more reliable performance evaluation. The main reported metrics are precision, recall, mAP50, and mAP50-95. mAP50 measures the detector's average precision, with a detection counted as correct when the intersection-over-union between predicted and ground-truth boxes exceeds 50%. mAP50-95 offers a stricter evaluation, averaging performance across IoU thresholds from 0.50 to 0.95. During training, the model automatically saves the best-performing weights (`best.pt`), selected by the highest validation mAP50. This final model is used within the system to generate bounding boxes over suspicious tumor regions.

8. Results

8.1 Multiclass classification

The primary evaluation used the combined test set and all three trained classifiers. Results show strong performance across all three, with accuracy and macro F1 above 98.7% in every case — indicating the selected architectures consistently differentiate between the project's four classes: glioma, meningioma, no tumor, and pituitary tumor.

Model	Accuracy	Balanced accuracy	Latency	Parameters
ConvNeXt-Tiny	99.29%	99.25%	5.3 ms	27.8 M
EfficientNet-B3	99.48%	99.48%	15.6 ms	10.7 M
ResNet-50	99.34%	99.30%	7.4 ms	23.5 M

Table 3. Per-model test performance and single-pass inference latency, as measured in the original evaluation environment (GPU). ConvNeXt-Tiny is fastest, EfficientNet-B3 is most accurate and most compact, ResNet-50 sits in between. Note: the live demo application (CPU-only, see the project README) reports different absolute latency numbers due to the hardware difference; relative model ranking is consistent.

Among the three, EfficientNet-B3 achieves the best overall performance — 99.48% accuracy and 99.46% macro F1 — making it the most balanced model in the system: it combines the highest accuracy with very high macro precision and macro recall. ResNet-50 performs nearly as well, and ConvNeXt-Tiny, while slightly lower, still delivers solid results. From a clinical perspective, the similarity across all three models suggests the processed dataset supports learning consistent patterns.

8.2 Per-class metrics of the best model

To examine behavior in more detail, per-class metrics were computed for EfficientNet-B3, the strongest overall model. Results are well balanced: No Tumor achieves the best F1-score (99.76%), followed by Pituitary (99.68%), Glioma (99.33%), and Meningioma (99.06%). Although meningioma has the lowest value, the gap relative to the other classes is small, so no significant performance degradation is observed in any category. The especially high recall on the Pituitary class indicates the model correctly identifies most cases in that class, while the 99.84% precision on No Tumor shows the system produces few false positives when classifying tumor-free images. Together, these metrics reinforce that EfficientNet-B3 not only achieves high accuracy overall, but maintains stable performance across classes with different sample counts.

8.3 Ensemble voting

The final class is not decided by a single model in isolation: the deployed system always soft-votes — averaging the softmax outputs of all three models and selecting the resulting top class.

99.48% SOFT-VOTE ACCURACY	99.46% MACRO F1	99.39% 3-WAY AGREEMENT
+5 net VOTING CORRECTION (+8 / -3)		

Voting corrects 8 test-set cases where at least one individual model was wrong, and only overturns 3 cases where a single model had been right — a net gain of +5 over picking any one model alone.

Class	Precision	Recall	F1
Glioma	99.50%	99.17%	99.33%
Meningioma	98.82%	99.29%	99.06%
No Tumor	99.84%	99.68%	99.76%
Pituitary	99.58%	99.79%	99.68%

Table 4. Per-class precision / recall / F1 for the soft-vote ensemble on the locked 2,114-image test set. No Tumor reaches the best F1 (99.76%) with very few false positives; Meningioma is lowest, at 99.06%.

Actual ↓ / Predicted →	Glioma	Meningioma	No Tumor	Pituitary
Glioma	596	4	1	0
Meningioma	1	420	0	2
No Tumor	2	0	616	0
Pituitary	0	1	0	471

Table 5. Soft-vote ensemble confusion matrix, 2,114-image locked test set. Diagonal = correct predictions. Only 11 errors total.

8.4 YOLO11n detector results

Beyond overall image classification, the system incorporates a supervised localization module based on YOLO11n, complementing the classifier's prediction with a spatial reference for the suspicious region.

Metric	Value	Meaning
mAP50	0.918	Detection counted correct at IoU > 0.50
mAP50-95	0.563	Stricter — averaged across IoU 0.50 → 0.95

Precision	0.907	Of the boxes drawn, how many are right
Recall	0.828	Of the real tumors, how many are caught

Table 6. YOLO11n evaluation on the patient-disjoint validation split of the Cheng dataset.

The YOLO11n detector reaches an mAP₅₀ of 0.918, indicating strong performance at general localization of tumor regions under a 50% IoU threshold. Precision of 0.907 shows a high proportion of the model's detections are correct, while a recall of 0.828 indicates there is still room to improve detection of all positive cases. The gap between mAP₅₀ and mAP₅₀₋₉₅ is notable: while mAP₅₀ evaluates localization under a comparatively lenient criterion, mAP₅₀₋₉₅ demands tighter alignment between predicted and ground-truth boxes. The 0.563 mAP₅₀₋₉₅ value suggests the detector correctly localizes tumor regions in general, but spatial precision could still be improved. YOLO11n therefore functions as a useful visual support module, though it should not be interpreted as a fully-validated clinical segmentation.

8.5 Efficiency vs. accuracy comparison

Beyond predictive performance, efficiency-related aspects were also evaluated — latency under Test-Time Augmentation, throughput, and model size. This matters because an interactive system must not only be accurate, but also respond in reasonable time. Results show ConvNeXt-Tiny is the fastest model (~5.3 ms latency, ~100 images/second throughput), but also the largest at 106 MB. EfficientNet-B3, despite somewhat higher latency, offers the best predictive performance and the smallest size among the three models evaluated, at 41 MB.

This comparison shows that the choice of primary model does not depend on speed alone. Although ConvNeXt-Tiny is more efficient at inference time, EfficientNet-B3 offers the best balance of accuracy, size, and performance stability, justifying its selection as the system's standout model. ResNet-50 maintains strong performance, though slightly below in accuracy and macro F1. The adoption of ConvNeXt-Tiny in the latest version, replacing a prior DenseNet-169 backbone, brings a substantial speed improvement without sacrificing overall accuracy.

8.6 Comparison with the prior binary project

Finally, the current system was compared against the earlier binary classification project. This comparison highlights the project's evolution from a solution focused only on differentiating tumor presence from absence toward a more complete, interpretable, and interactive multiclass system. The comparison shows the new system is not simply an expansion of the number of classes, but a structural evolution: while the binary system focused on detecting tumor presence or absence, the multiclass system adds tumor-type-specific classification, uncertainty estimation, out-of-distribution detection, visual explainability, supervised localization, and clinical narrative generation.

8.7 Global interpretation of results

Overall, the results indicate the developed system achieves strong performance at multiclass brain tumor classification in MRI. EfficientNet-B3 stands out as the strongest model, with 99.48% accuracy and 99.46% macro F1, while maintaining balanced behavior across the four evaluated classes. The addition of YOLO11n complements classification with visual localization of the suspicious region, while the uncertainty, OOD-detection, and visual-interpretability modules add information about the reliability and reviewability of the result. The comparison with the previous binary project shows the current system significantly extends the functional scope of the work, moving from a binary prediction to a more complete, modular solution. These results, however, should be interpreted within an experimental framework: although the metrics are high, any real clinical application would require external validation with data from multiple centers, evaluation by specialists, and prospective studies confirming the system's robustness in real usage scenarios.

9. Conclusions

The brAIn multiclass project demonstrates the feasibility of building an experimental support system for brain MRI analysis in a local environment, using conventional computing hardware and a maintainable, modular architecture. The developed system integrates a complete flow, from image upload to visual and narrative result generation, combining multiclass classification, uncertainty estimation, out-of- distribution detection, visual interpretability, supervised tumor localization, and clinical narrative generation.

Results show strong performance at the multiclass classification task, with values above 99% accuracy and macro F1 for the best-evaluated model. The system also incorporates elements that extend beyond a conventional classifier — uncertainty measurement via MC Dropout, OOD detection, supervised localization with a YOLO11n detector trained on an independent radiological dataset, hierarchical heat-map generation, and a frontend interface oriented toward interactive, visual review of results.

Beyond the quantitative metrics, one of the project's main contributions is its integrative approach. Combining specialized components — convolutional neural networks, object detectors, segmentation models, and language models — allows for a more complete system than the sum of its individual parts. At the same time, development also highlighted the importance of respecting each module's functional boundaries: language models are useful for generating structured textual explanations, but should not replace supervised modules when a task requires precise spatial localization or decisions based directly on the image.

The project should therefore not be understood as an autonomous clinical diagnostic tool, but as a solid experimental foundation for exploring how different AI techniques can be combined into a more interpretable, reviewable support system oriented toward medical image analysis.

10. Future Work

10.1 Dataset improvement and external validation

One of the main lines of improvement is to expand the diversity and quality of the data used. Future versions should incorporate sets such as BraTS 2023/2024, anonymized hospital data, and studies with multiple MRI sequences (T1, T1c, T2, FLAIR). This would overcome the current limitation of relying mainly on axial slices and improve the system's ability to generalize across acquisition types and institutions. A more rigorous cross-dataset validation would also be needed, reporting metrics separately by data origin — e.g. Kaggle, BRISC, Cheng, or potential hospital data — to detect domain-shift effects, i.e. performance variation when the model is applied to images from sources different from those used in training.

10.2 Clinical validation and specialist evaluation

Another fundamental improvement would be prospective validation with radiologists or neuroimaging specialists, comparing the system's predictions against expert assessment, measuring agreement between the two, and analyzing in which cases the tool provides real value as a support system. This validation would also help determine whether the system's outputs are understandable, useful, and clinically relevant — an essential step before considering any integration into real medical environments.

10.3 Improved tumor localization and segmentation

Although the current system uses YOLO11n and MobileSAM to complement classification with visual localization, a future direction would be to replace or reinforce MobileSAM with models more specialized for medical imaging, such as MedSAM — in particular, exploring LoRA fine-tuning on the Cheng dataset to obtain masks better adapted to brain MRI. This would allow moving from bounding-box or approximate-mask localization toward more precise tumor segmentation, evaluated with segmentation-specific metrics such as Dice, IoU, or Hausdorff Distance, compared against expert annotations.

10.4 Automatic MRI sequence detection

Another relevant improvement would be an automatic MRI-sequence detector able to identify whether an image corresponds to T1, T1c, T2, or FLAIR — warning the user when the uploaded sequence is not the most appropriate for the expected analysis, or does not match the system's primary training domain. This would be especially important in a real deployment, since different MRI sequences carry complementary information and can change the tumor's visual appearance; correctly identifying the input modality would improve the reliability and traceability of the analysis.

10.5 Clinical narrative and multimodal models

For clinical text generation, a future direction would be to systematically compare MedGemma 1.5 4B against other multimodal or language models, such as Qwen3-VL or other specialized VLMs, to determine whether the current model is the most suitable for generating clinical narrative or whether alternatives offer greater accuracy, stability, or communicative clarity.

10.6 Interface and user-experience improvements

From a functional standpoint, a useful improvement would be automatic PDF report export — generating a structured one-page document with the analyzed image, bounding box, orientative score, MedGemma-generated narrative, and user-selected symptoms. An interactive glossary with plain-language explanations of complex medical terms would also help non-specialist users understand the reports. Another possible addition is a local analysis history, allowing several scans of the same case to be compared over time and showing the evolution of estimated size — though this would need to be designed carefully to protect data privacy.

10.7 Privacy, security, and deployment

Finally, an important line of improvement is strengthening the system's privacy and security guarantees. Although the current version runs locally and does not send images to external servers, a future adaptation to a clinical environment should include encrypted communications, secure storage, automatic deletion of images after analysis, and access-control mechanisms if multiple users are supported. Since the project sits in a medical context, any real deployment should consider applicable data-protection requirements — GDPR in the European context, and HIPAA if a U.S. environment were considered. This adaptation would be necessary to evolve from a local prototype into a tool with stronger guarantees for use in professional environments.

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Appendix — Interface Walkthrough

The figures throughout Section 6 above are drawn from this appendix set, captured directly from the running application. For a complete, click-through walkthrough of every tab with additional screenshots, see the project's demo notebook (`notebooks/demo.ipynb`) in the GitHub repository, which also covers preprocessing, the full augmentation breakdown, and the bilingual (EN/ES) report generation shown in both languages.